

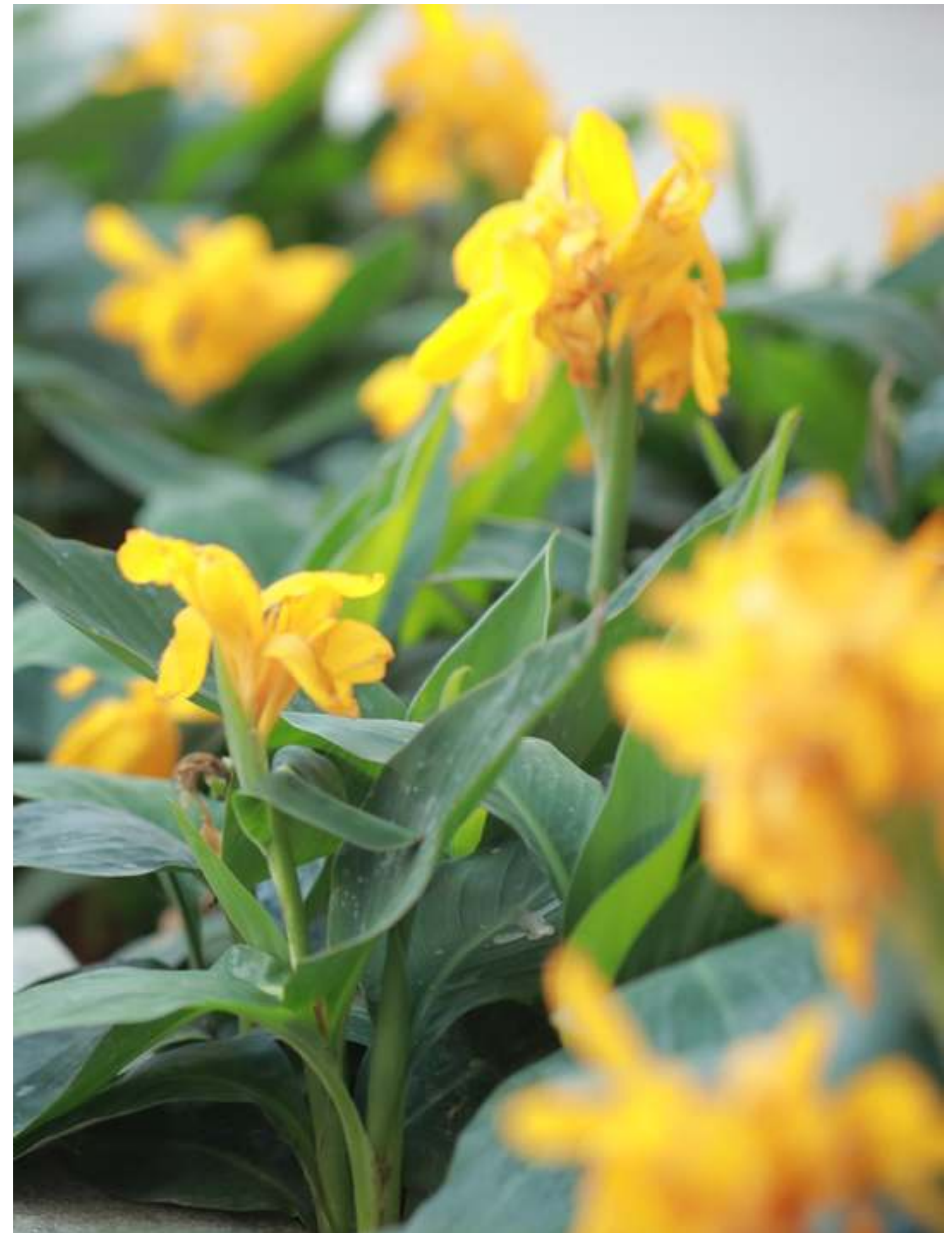
MACHINE LEARNING FOR SIGNAL PROCESSING

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<http://leap.ee.iisc.ac.in/sriram/teaching/MLSP25/>



VECTOR CALCULUS

➤ Let $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ denote a real n dimensional vector $\in \mathcal{R}^{n \times 1}$ and let $f(\mathbf{x})$ denote a scalar function which maps $\mathcal{R}^n \rightarrow \mathcal{R}$

➤ Then, we define $\frac{\partial f}{\partial \mathbf{x}} = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix}$ is derivate of the function $f(\mathbf{x})$ w.r.t. $\mathbf{x}$

➤ Question :

- * What is the derivative of $f(\mathbf{x}) = \mathbf{a}^T \mathbf{x}$
- * What is the derivative of $f(\mathbf{x}) = \mathbf{x}^T \mathbf{x}$

VECTOR DERIVATIVES

Let $\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ denote a real n dimensional vector $\in \mathcal{R}^{n \times 1}$ and let $\mathbf{f}(\mathbf{x})$ denote a vector function which maps $\mathcal{R}^n \rightarrow \mathcal{R}^m$

... Let $\mathbf{f}(\mathbf{x}) = \begin{bmatrix} f_1(\mathbf{x}) \\ f_2(\mathbf{x}) \\ \vdots \\ f_m(\mathbf{x}) \end{bmatrix}$

Then, we define $\frac{\partial \mathbf{f}}{\partial \mathbf{x}} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \frac{\partial f_m}{\partial x_2} & \cdots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}$ is derivate of the function $\mathbf{f}(\mathbf{x})$ w.r.t. $\mathbf{x}$... The derivative $\in \mathcal{R}^{m \times n}$

MATRIX DERIVATIVES

➤ Let $\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$ denote a real $m \times n$ matrix and let $f(\mathbf{A})$ denote a scalar function which maps $\mathcal{R}^{m \times n} \rightarrow \mathcal{R}$

➤ Then, we define $\frac{\partial f}{\partial \mathbf{A}} = \begin{bmatrix} \frac{\partial f}{\partial a_{11}} & \frac{\partial f}{\partial a_{12}} & \cdots & \frac{\partial f}{\partial a_{1n}} \\ \frac{\partial f}{\partial a_{21}} & \frac{\partial f}{\partial a_{22}} & \cdots & \frac{\partial f}{\partial a_{2n}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial f}{\partial a_{m1}} & \frac{\partial f}{\partial a_{m2}} & \cdots & \frac{\partial f}{\partial a_{mn}} \end{bmatrix}$ is derivative of the function $f(\mathbf{A})$ w.r.t. $\mathbf{A} \dots$

➤ This derivative $\in \mathcal{R}^{m \times n}$

MATRIX DERIVATIVE PROBLEMS

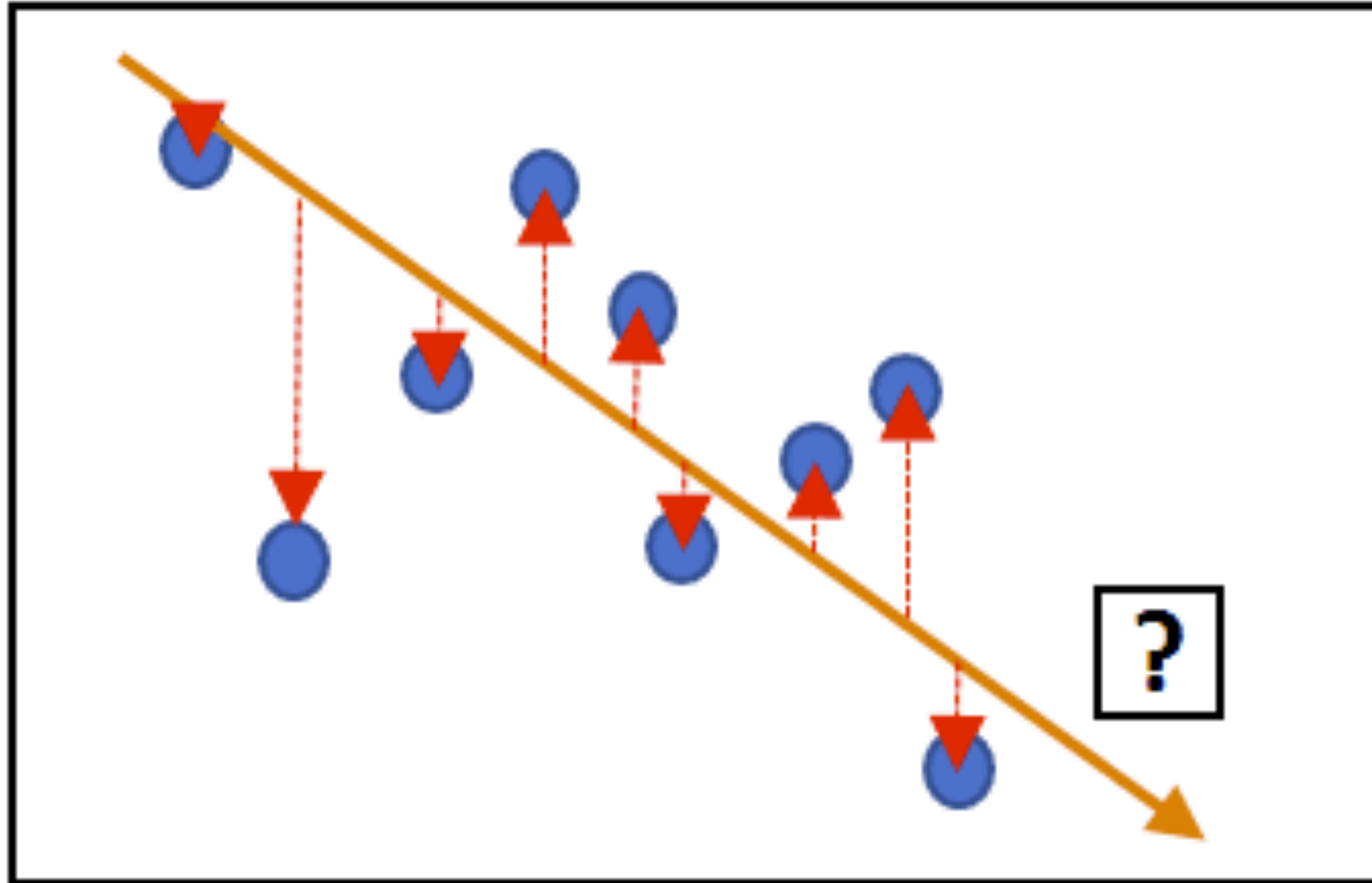
➤ Question :

- * What is the derivative of $f(\mathbf{x}) = \mathbf{x}^T \mathbf{A} \mathbf{x}$ w.r.t $\mathbf{x}$
- * What is the derivative of $f(\mathbf{A}) = \mathbf{x}^T \mathbf{A} \mathbf{x}$ w.r.t $\mathbf{A}$
- * What is the derivative of $f(\mathbf{A}) = \text{Tr}(\mathbf{A})$ w.r.t $\mathbf{A}$
- * What is the derivative of $f(\mathbf{A}) = \text{Tr}(\mathbf{A}\mathbf{B})$ w.r.t $\mathbf{A}$
- * What is the derivative of $f(\mathbf{A}) = \text{Tr}(\mathbf{A}\mathbf{B}\mathbf{A}^T)$ w.r.t $\mathbf{A}$
- * What is the derivative of $f(\mathbf{A}) = \log |\mathbf{A}|$ w.r.t $\mathbf{A}$

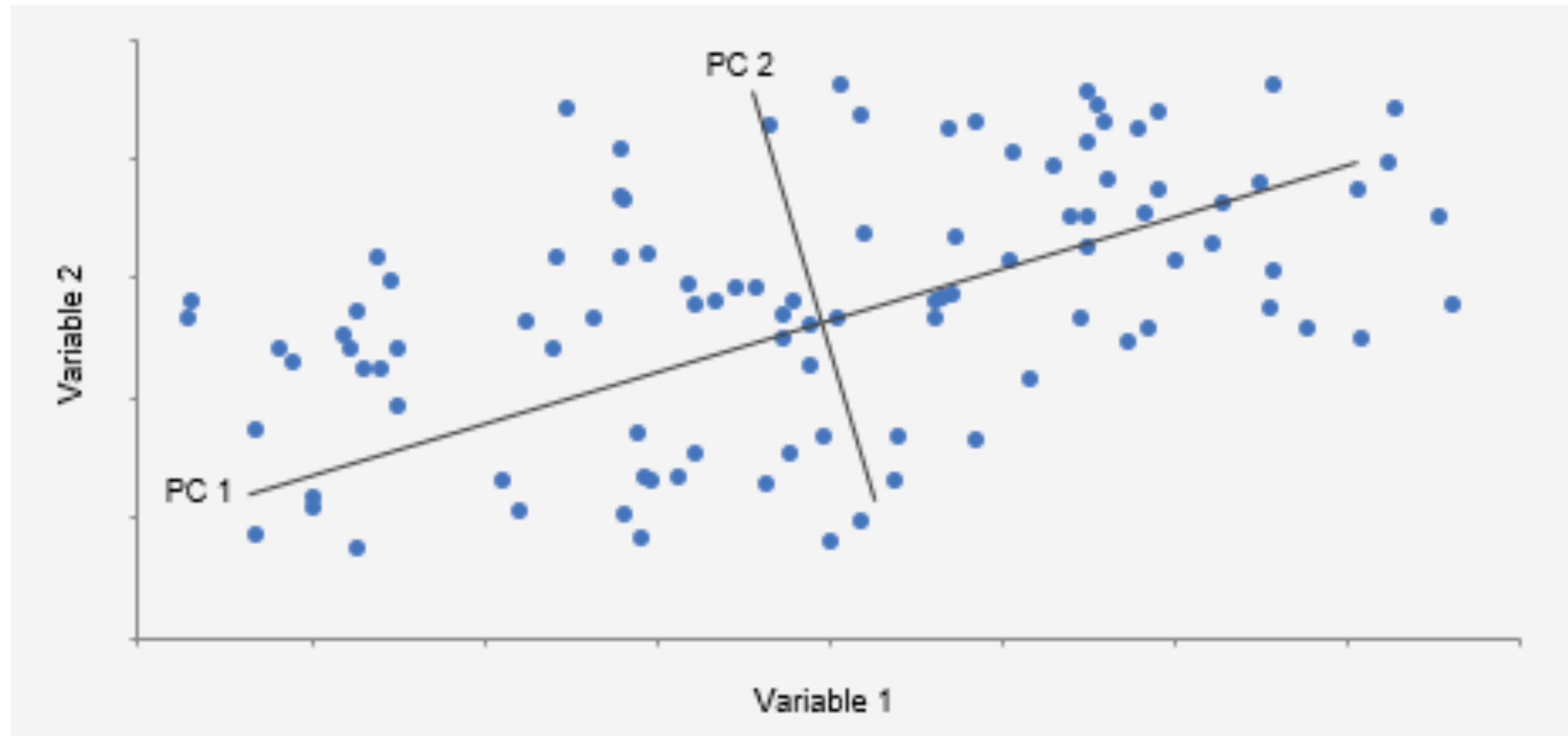
PRINCIPAL COMPONENT ANALYSIS

- ❖ Reducing the data $\mathbf{x}_n$ of dimension D to lower dimension
- ❖ Projecting the data into subspace which preserves maximum data variance
- ✓ Maximize variance in projected space $M < D$
- ❖ Equivalent formulated as minimizing the error between the original and projected data points.

DIRECTION OF MAXIMUM VARIANCE



PCA EXAMPLE



PRINCIPAL COMPONENT ANALYSIS

- ❖ First M eigenvectors of data covariance matrix

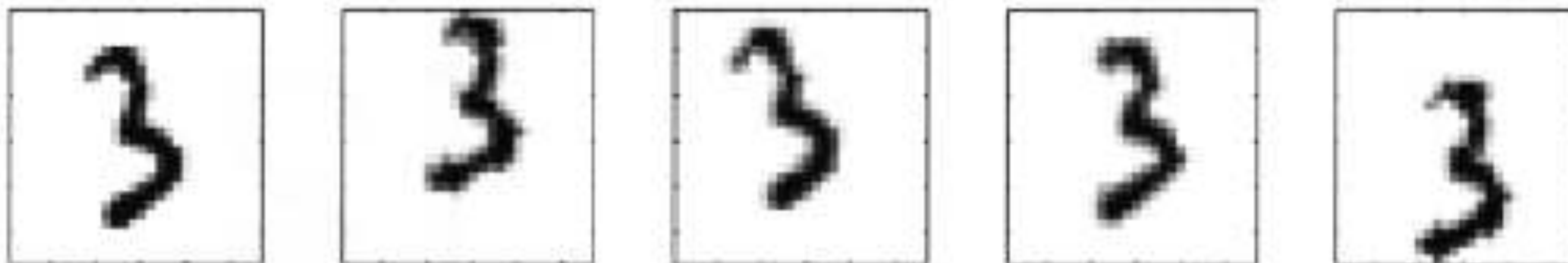
$$S = \frac{1}{N} \sum_{n=1}^N (\mathbf{x}_n - \bar{\mathbf{x}})(\mathbf{x}_n - \bar{\mathbf{x}})^T$$

- ❖ Residual error from PCA

$$J = \sum_{i=M+1}^D \lambda_i$$

PRML - C. Bishop (Sec. 12.1)

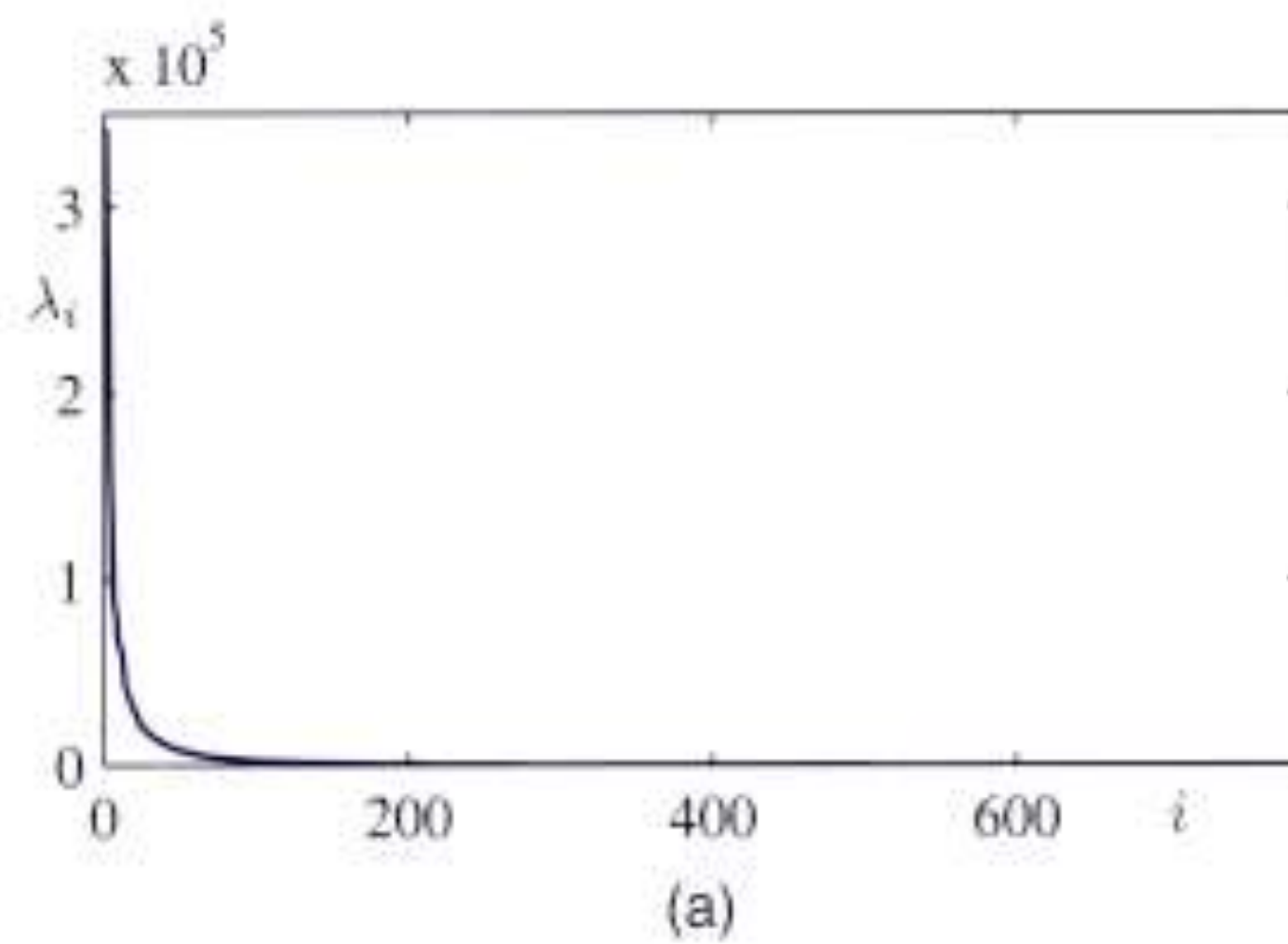
- ❖ First eigenvectors of data covariance matrix



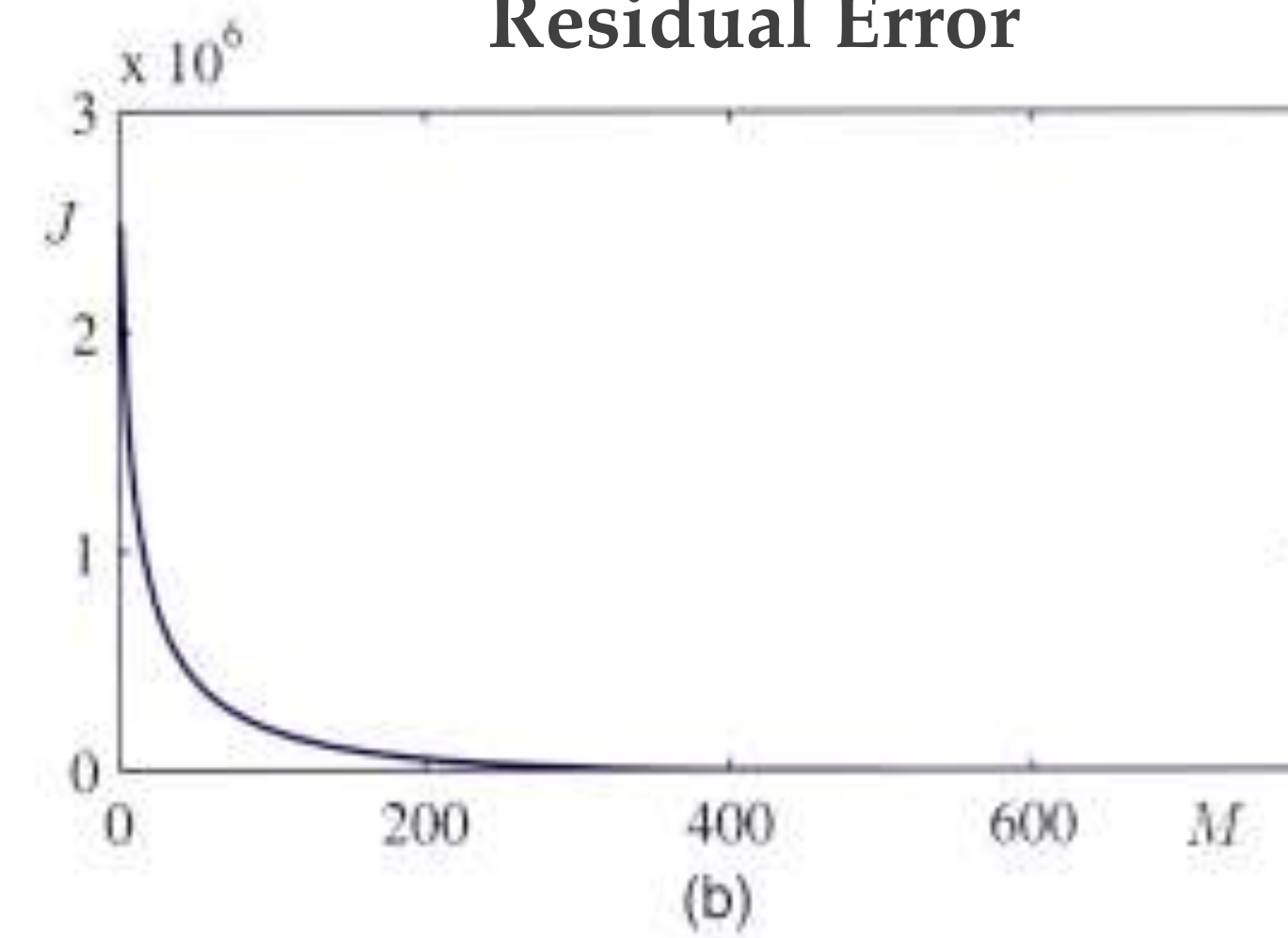
- ❖ Residual error from PCA

Handwritten digits used for PCA training...

Eigen Values

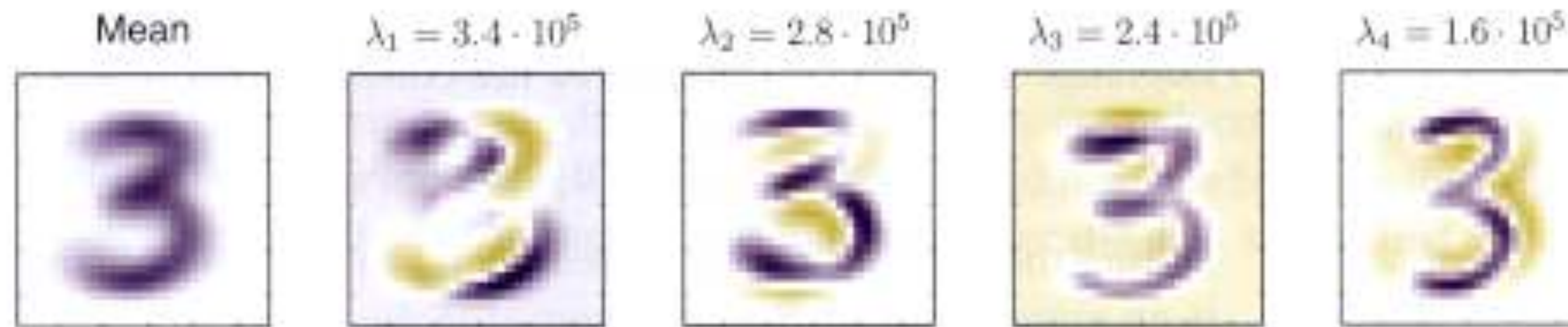


Residual Error

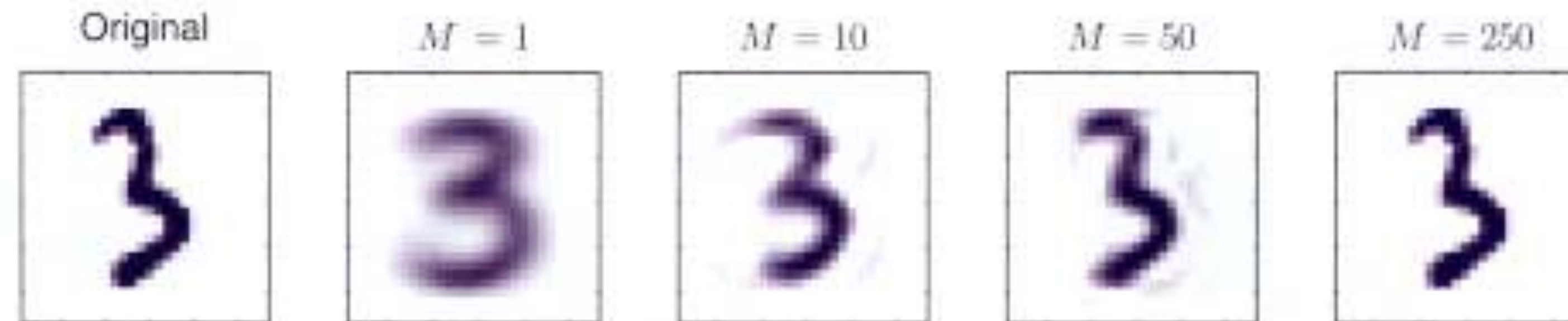


PCA - RECONSTRUCTION

Eigenvectors

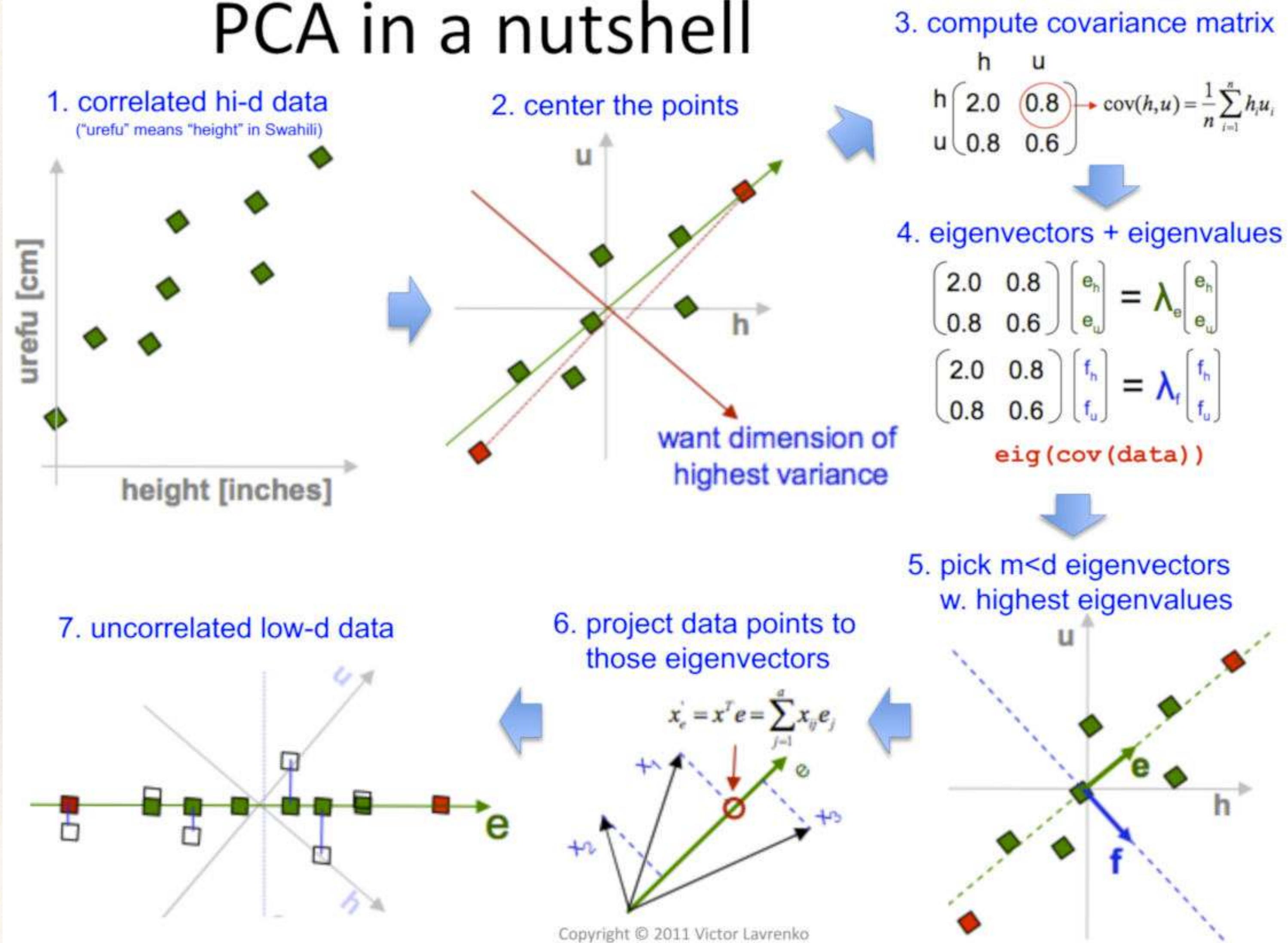


PCA - Reconstruction

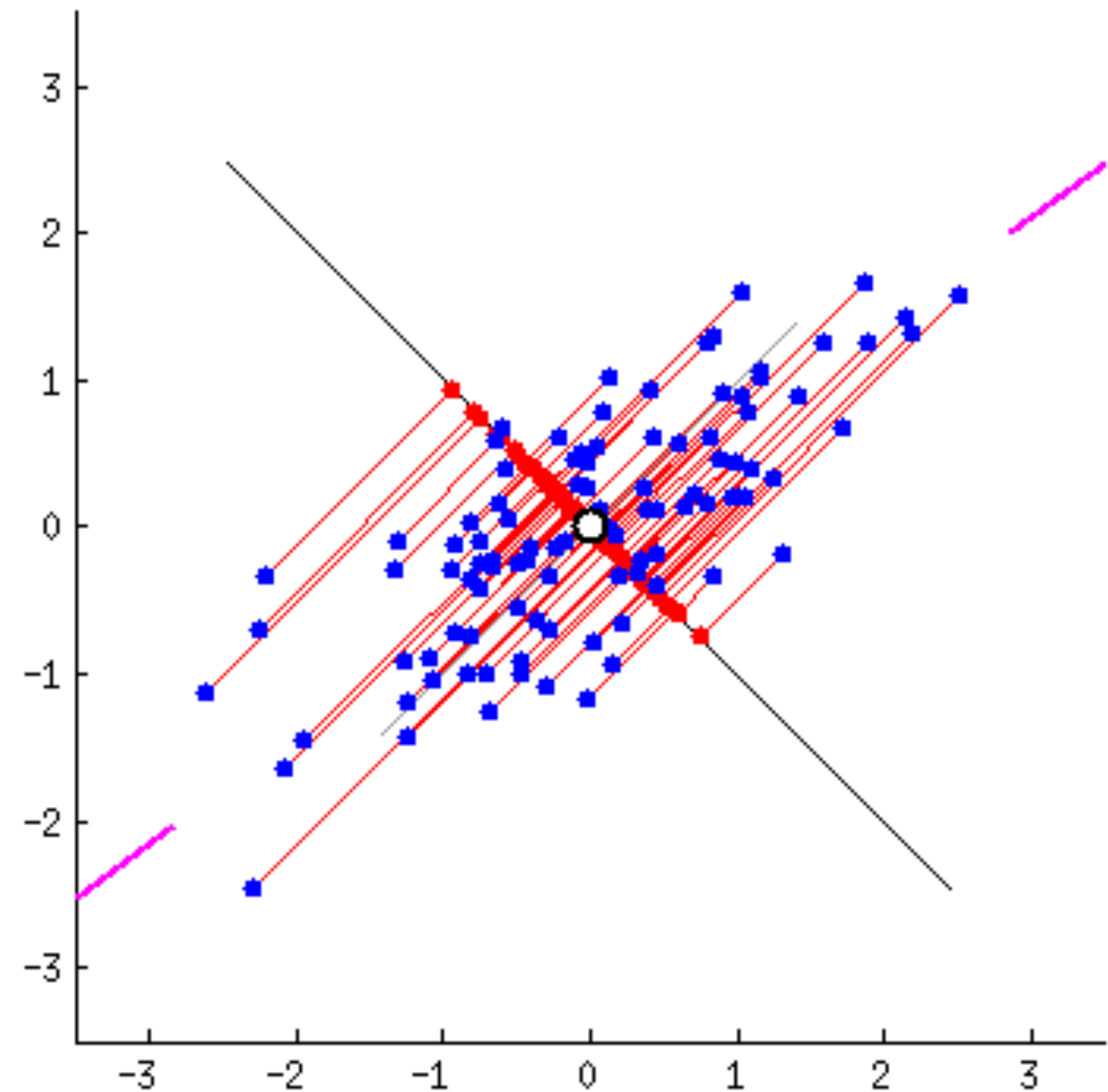
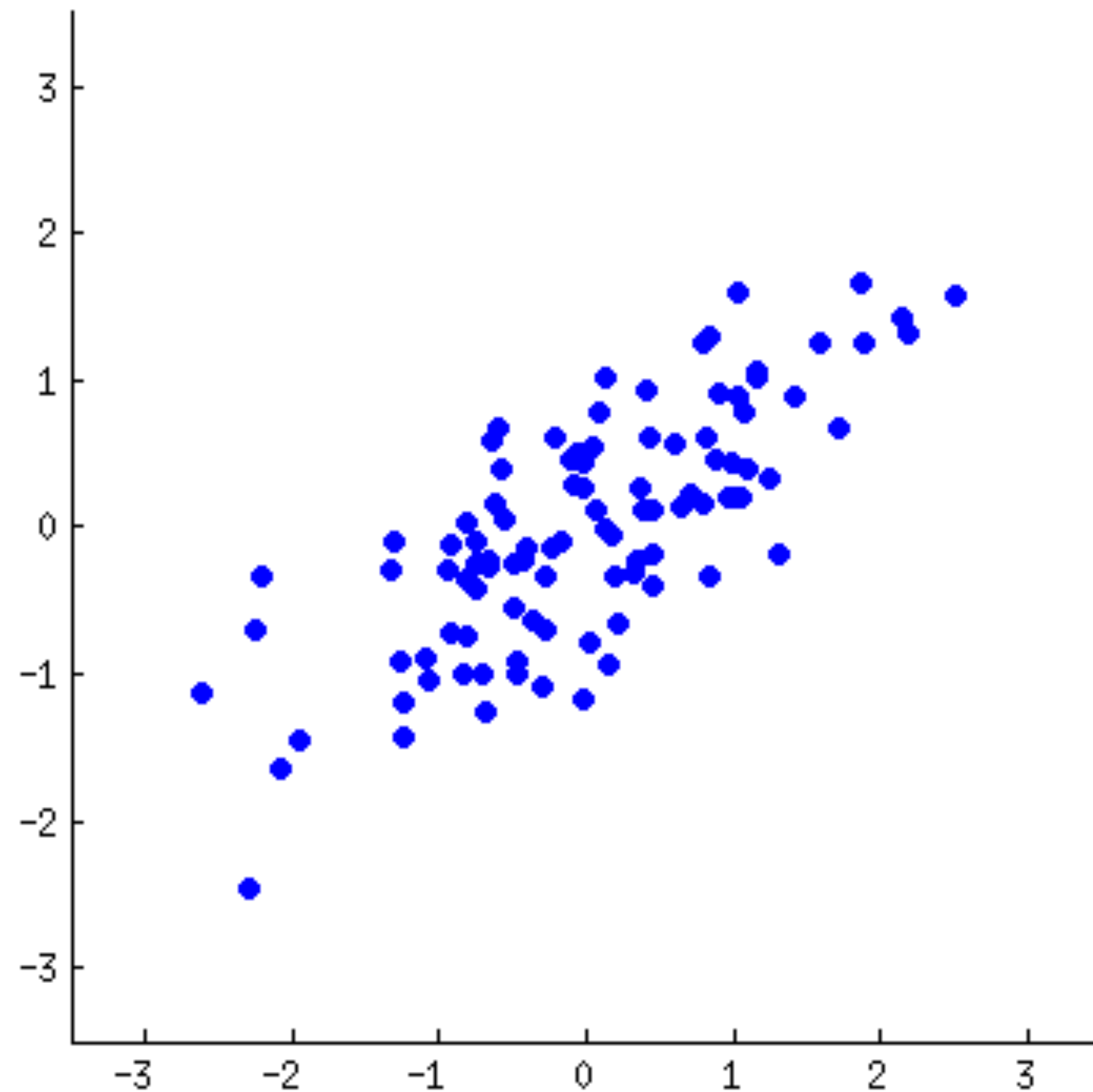


PCA Summary

PCA in a nutshell

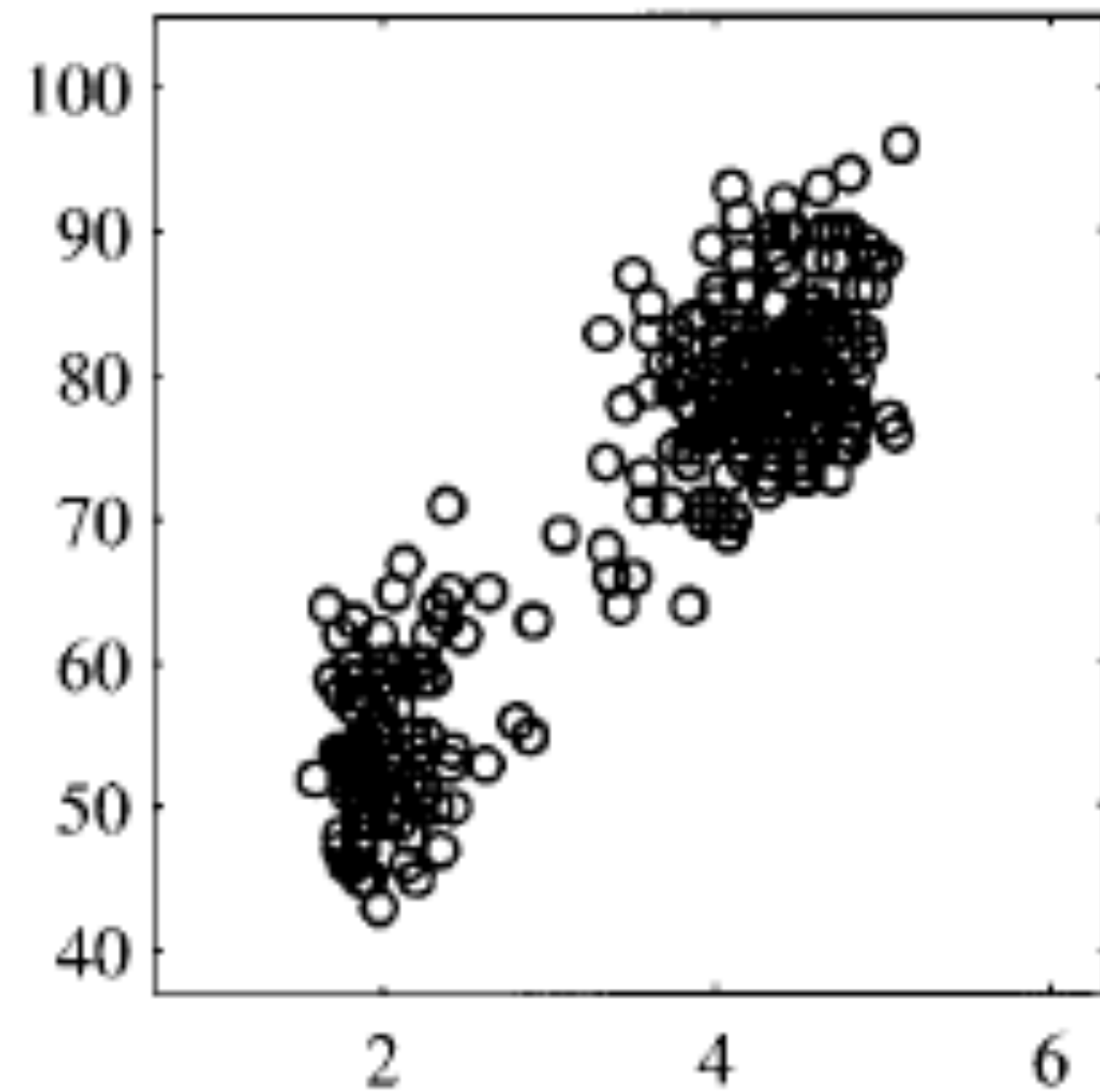


Visualizing PCA

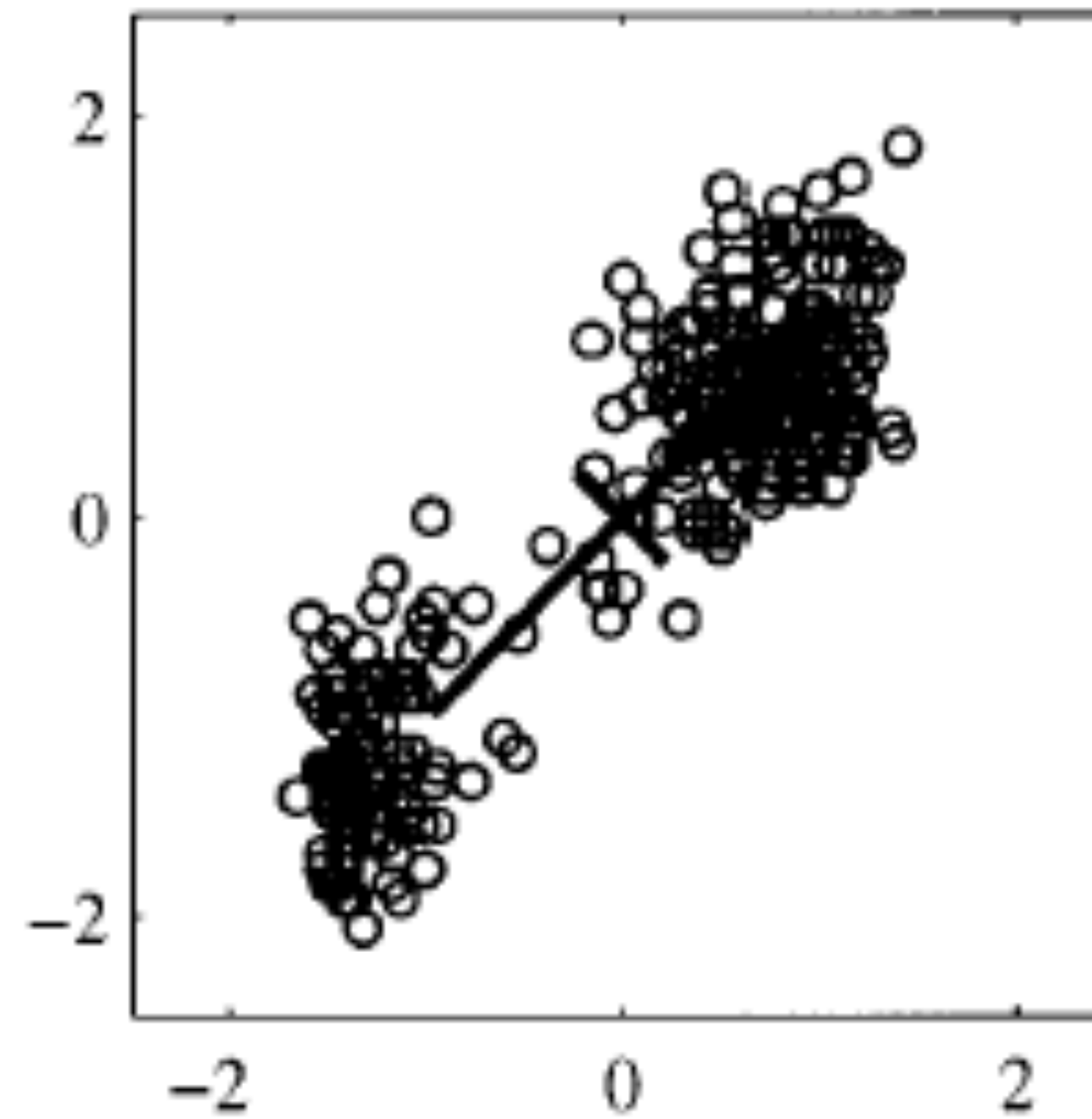


WHITENING THE DATA

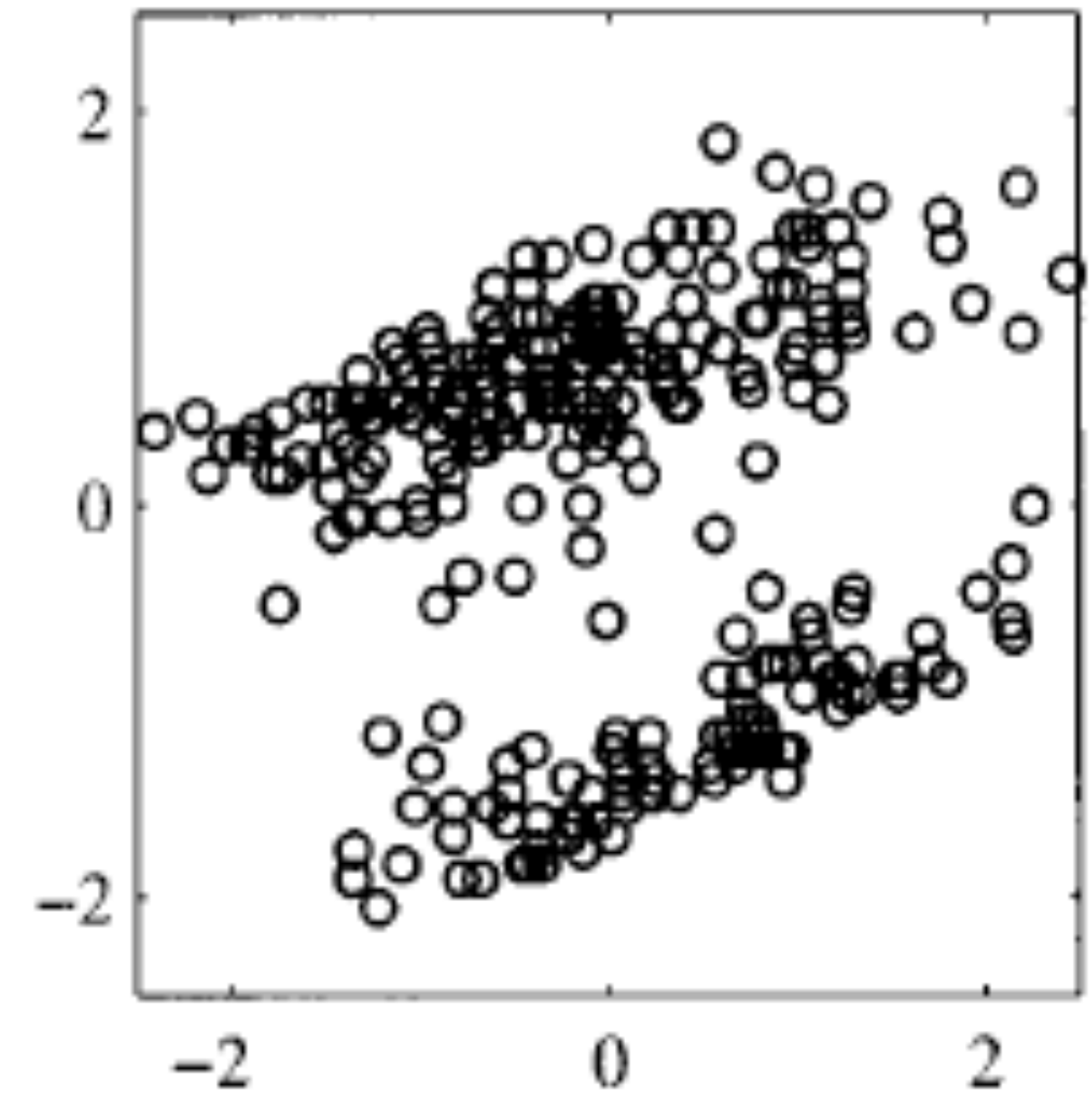
Original Data



Standardization



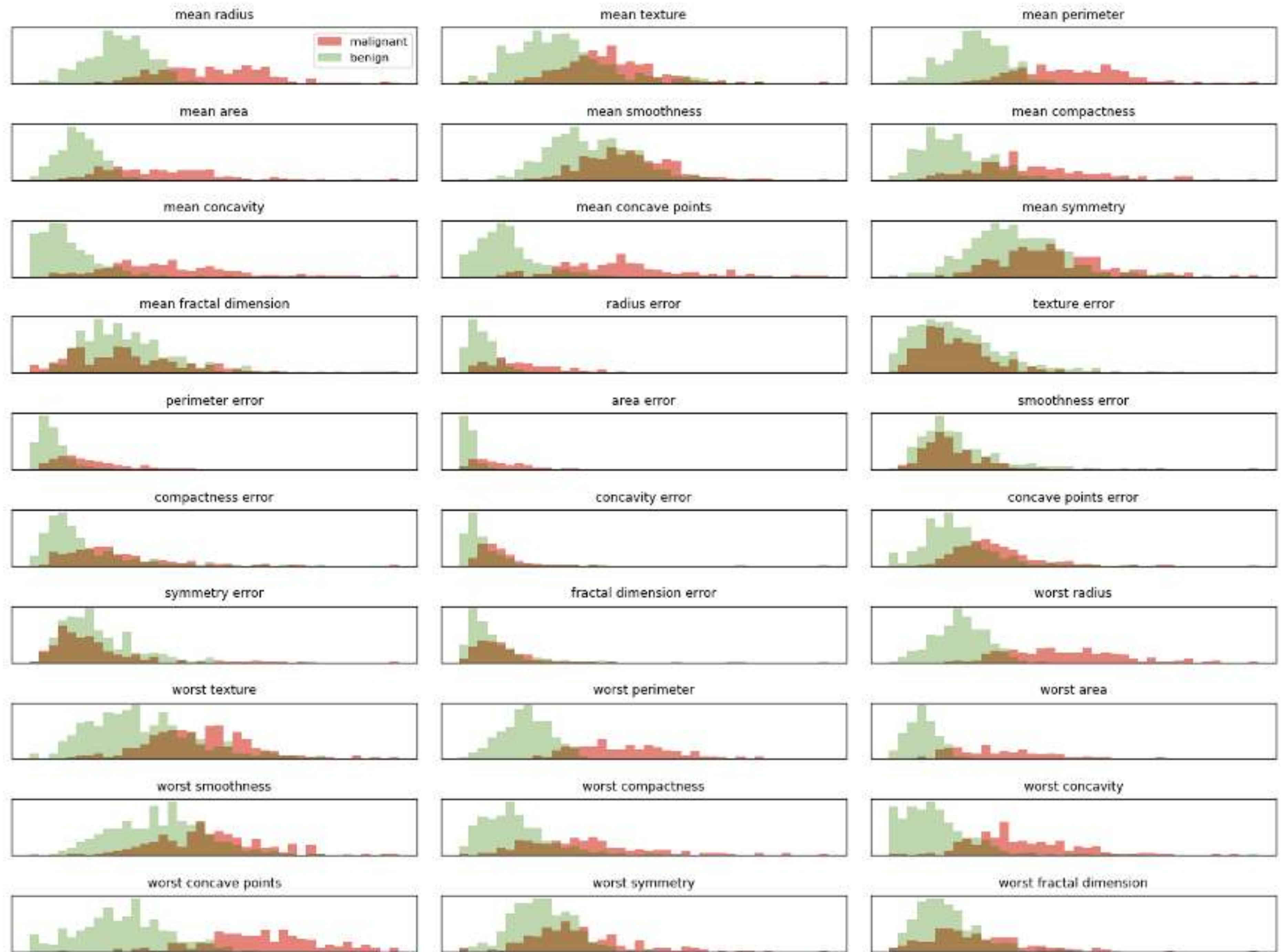
Whitening



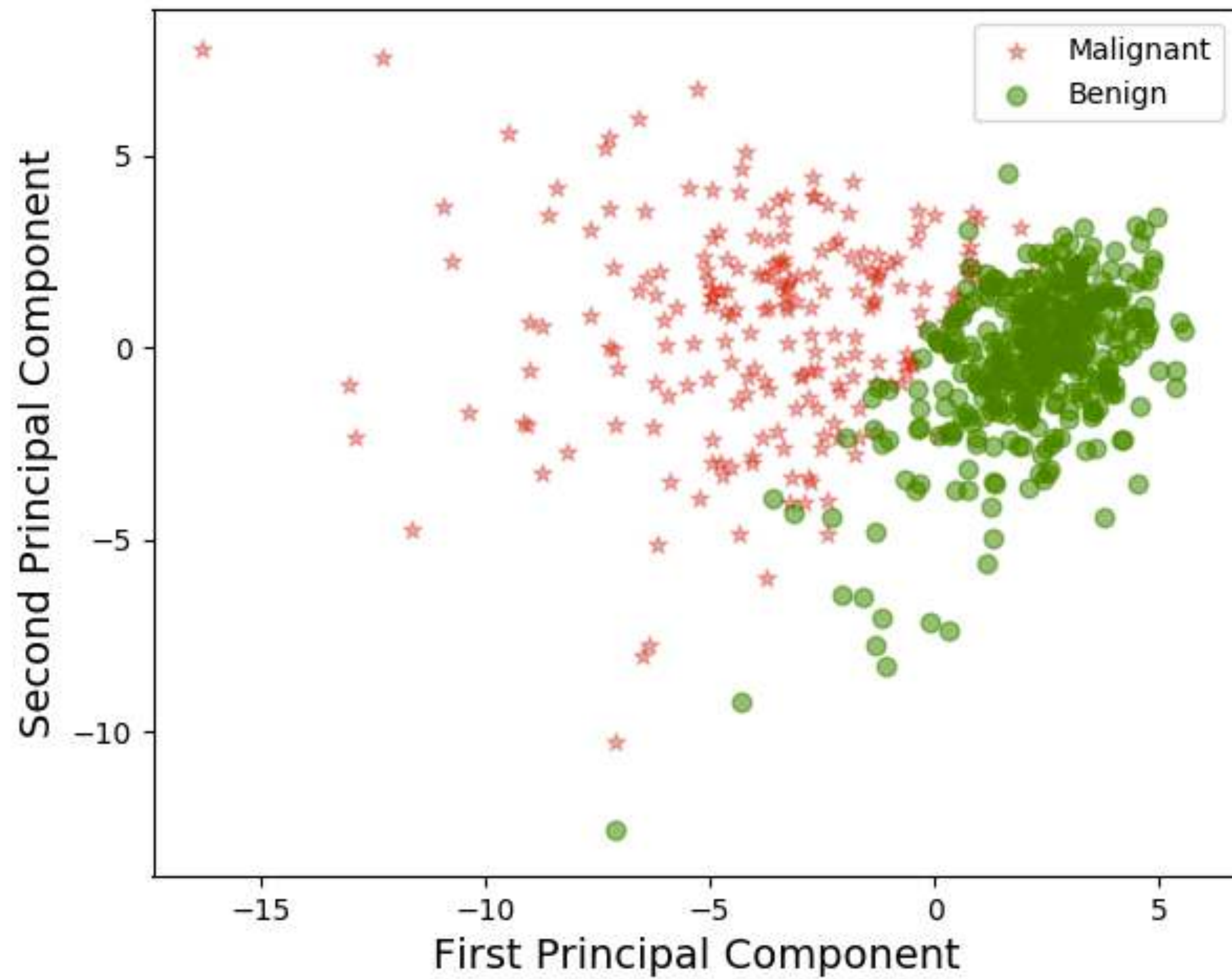
APPLICATION

- ❖ Wisconsin Cancer dataset ([https://archive.ics.uci.edu/ml/datasets/Breast+Cancer+Wisconsin+\(Diagnostic\)](https://archive.ics.uci.edu/ml/datasets/Breast+Cancer+Wisconsin+(Diagnostic)))
- ❖ 569 participants
- ❖ 212 (M) 357 (B)
- ❖ 30 features —> digitized image of a fine needle aspirate (FNA) of a breast mass. The features describe characteristics of the cell nuclei present in the image.

Raw Features



PCA



THANK YOU

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